

**An Internship Report**  
Of  
**Artificial Intelligence and Machine Learning**

Submitted  
To

**School of Computer Science and Engineering**  
**IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of  
**B.Tech CSE(Computer Science and Engineering)**



By

Roll Number of Student: 2410031176  
Name of Student: ARVINDER SINGH BAGGA  
Batch: 2024-2028

## **CANDIDATE'S DECLARATION**

I, ARVINDER SINGH BAGGA, do hereby declare that the internship report titled “Artificial Intelligence and Machine Learning” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source.

I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Arvinder Singh Bagga

Roll No.: 2410031176

## **ACKNOWLEDGEMENT**

I would like to express my sincere gratitude to everyone who supported and guided me throughout my Artificial Intelligence and Machine Learning Internship.

I am thankful to Kodacy Technologies and the AICTE Virtual Internship Programme for providing me with the opportunity to explore and learn various concepts related to Artificial Intelligence.

This internship provided me a great exposure to theory as well as the applications of Artificial Intelligence and Machine Learning. The practical experience of working with modern ideas and technologies of artificial intelligence and machine learning that they provided helped me understand key topics like machine learning algorithms, deep learning architectures, data analysis, model development, validation, and optimization.

I would also like to express my sincere gratitude to my faculty members and IILM University, Greater Noida, for their guidance and support. Finally, I am deeply thankful to my parents and everyone who encouraged me throughout this learning journey.

Signature:  
Student Name: Arvinder Singh Bagga  
Roll No : 2410031176

## CONTENTS

| Chapter No. | Chapter Name                      | Page Nos. |
|-------------|-----------------------------------|-----------|
|             | Cover Page                        | ----      |
| 1.          | Candidate's Declaration           | i         |
| 2.          | Acknowledgment                    | ii        |
| 3.          | Internship Completion Certificate | iii       |
| 4.          | Project Description               | ---       |
|             | 4.1 Introduction                  | 1         |
|             | 4.2 Organization Profile          | 2         |
|             | 4.3 Problem Statement             | 2         |
|             | 4.4 Project Objectives            | 3         |
|             | 4.5 Scope of the Project          | 3         |
|             | 4.6 Technologies and Tools Used   | 4         |
|             | 4.7 System Architecture           | 5         |
|             | 4.8 Methodology                   | 5         |
|             | 4.9 Expected Outcomes             | 6         |
|             | 4.10 Certificate of Completion    | 6         |
| 5.          | Bibliography/References           | 7         |

# INTERNSHIP COMPLETION CERTIFICATE

Kodacy

SPACE  
Scientific Platforms And Cosmic Explorations



## Certificate OF INTERNSHIP

This certificate is presented to:

**Arvinder Singh Bagga**

for successfully completing the 30 day virtual internship programme on

**Artificial Intelligence and Machine Learning**

conducted by KODACY in association with Scientific  
Platforms And Cosmic Explorations (SPACE).



Date Of Completion : September 8, 2026  
Certificate ID : 644b2875e2f9e315  
< Scan this QR to check validity



*Akash Joseph*  
Authorized Signature  
Akash Joseph (CEO)

## **CHAPTER 4: PROJECT DESCRIPTION**

### **4.1 Introduction**

The internship in Artificial Intelligence and Machine Learning was a valuable learning experience that helped me understand how intelligent systems and data-driven technologies work in practice enabling machines to perform tasks that traditionally required human intelligence, such as learning, reasoning, problem-solving, perception and language understanding. As part of my academic curriculum for the B.Tech Computer Science and Engineering programme at IILM University, Greater Noida, I undertook an AICTE-recognised 30-day virtual internship on “Artificial Intelligence and Machine Learning” conducted by Kodacy in association with Scientific Platforms And Cosmic Explorations . This internship was designed to provide a strong foundation in the core concepts, techniques and real-world applications of AI and ML.

The internship covered foundational topics including the history and evolution of AI, types of AI (narrow, general and super AI), machine learning paradigms (supervised, unsupervised and reinforcement learning), neural networks, natural language processing, and the ethical considerations surrounding the deployment of AI systems. This report documents the learning journey, the organization under which the internship was completed, the objectives pursued, the tools and technologies explored, and the outcomes achieved during the internship period.

Overall, the internship gave me a broader understanding of the Artificial Intelligence and Machine Learning domain. It helped me connect the concepts learned academically with the practical steps involved in developing AI and Machine Learning systems.

## **4.2 Organization Profile**

The internship was completed under Kodacy, an education technology platform that focuses on redefining education by offering industry-relevant, skill-based virtual internship and certification programs to students across various technical domains. Kodacy provides AICTE Virtual Internship programs that allow students to gain practical exposure to emerging technologies such as Artificial Intelligence, Machine Learning, Data Science and Web Development, through structured online modules, assessments and mentorship.

Through the internship, I was able to explore how different areas of AI and Machine Learning are connected. The programme included both conceptual understanding and technical learning related to algorithms, data processing, neural networks, and Machine Learning workflows.

The virtual mode of the internship also made it possible to learn through structured modules and technical resources. Overall, the internship provided useful exposure to the AI and Machine Learning domain and helped me develop a better understanding of the technologies and processes involved in building intelligent systems.

## **4.3 Problem Statement**

Despite AI being a rapidly growing field with wide-ranging applications, many undergraduate students lack structured, practical experience or proper learning of basic AI concepts in their regular classes exposure to its fundamental concepts within a typical classroom environment . There exists a gap between theoretical knowledge taught in academic curriculum and the practical, applied understanding required by the industry. This internship aimed to address that gap by providing a structured, self-paced learning path covering the fundamentals of AI, along with practical exercises to reinforce conceptual understanding. A typical Machine Learning system involves several stages. It starts with understanding the problem and working with data, followed by data preparation, feature representation, model selection, training, evaluation, and improvement. Without understanding this complete workflow, it can be difficult to see how theoretical concepts are applied in practical situations.

#### **4.4 Project Objectives**

The main objective of this internship was to develop a strong understanding of the fundamental and practical concepts of Artificial Intelligence and Machine Learning.

The specific objectives of the internship were:

- To understand the fundamental concepts, history and scope of Artificial Intelligence.
- To study the different types and branches of AI, including Machine Learning and Deep Learning.
- To gain a working understanding of supervised, unsupervised and reinforcement learning techniques.
- To explore the basic architecture and working of neural networks.
- To understand real-world applications of AI across industries such as healthcare, finance, and automation.
- To become familiar with the ethical, social and safety considerations associated with AI systems.
- To gain knowledge of Deep Learning and Artificial Neural Networks.
- To study different machine learning algorithms and understand how they are used to solve real-world problems.
- To evaluate the accuracy and performance of the developed machine learning model
- To identify the advantages and limitations of AI and ML in different fields.
- To explore the future applications and development of AI and ML in education, healthcare, business, and other areas.

#### **4.5 Scope of the Project**

The scope of this internship was limited to building a conceptual and foundational understanding of Artificial Intelligence rather than developing a full-scale deployable AI product. It covered the theoretical underpinnings of AI and Machine Learning, an overview of commonly used algorithms and tools, and an introduction to how AI systems are designed, trained and evaluated. The internship also touched upon the



broader implications of AI adoption in society, industry and everyday life, providing a well-rounded introductory exposure to the field.

The internship began with the fundamentals of Artificial Intelligence. This included concepts related to reasoning, search techniques, knowledge representation, and intelligent problem-solving. These topics provided a foundation for understanding how AI systems approach and solve different types of problems.

Overall, the internship provided a broad introduction to the different areas of AI and Machine Learning and helped in understanding how these areas are connected.

#### **4.6 Technologies and Tools Used**

During the AI and Machine Learning internship, various technologies and tools were used to understand concepts, work with data, build machine learning models, and analyze results.

1. Python – Python was used as the main programming language for developing and testing AI and ML models.
2. Jupyter Notebook – Jupyter Notebook was used to write and run Python code, analyze data, and display results.
3. NumPy – NumPy was used for numerical calculations and working with arrays and mathematical operations.
4. Pandas – Pandas was used to collect, organize, clean, and analyze datasets.
5. Matplotlib – Matplotlib was used to create graphs and charts to understand data visually.
6. Seaborn – Seaborn was used to create attractive and easy-to-understand data visualizations.
7. Scikit-learn – Scikit-learn was used to build, train, test, and evaluate machine learning models.
8. Machine Learning Algorithms – Algorithms such as Linear Regression, Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbors were studied and used for different tasks.
9. Google Colab – Google Colab was used as an online platform for writing and running Python programs and machine learning experiments.
10. Git and GitHub – Git and GitHub were used to store, manage, and track changes in the project code and files.

#### **4.7 System Architecture**

A typical AI/Machine Learning system follows a structured pipeline. The general architecture studied during the internship can be summarized as follows:

- Data Collection – gathering raw data relevant to the problem statement.
- Data Pre-processing – cleaning, transforming and preparing data for analysis (handling missing values, normalization, encoding).
- Feature Selection/Extraction – identifying the most relevant attributes that influence the outcome.
- Model Selection and Training – choosing an appropriate algorithm (e.g. regression, classification, clustering) and training it on the prepared dataset.
- Model Evaluation – testing the trained model on unseen data using metrics such as accuracy, precision, recall and F1-score.
- Deployment and Monitoring – integrating the model into a real-world application and monitoring its performance over time.

#### **4.8 Methodology**

The internship followed a self-paced, module-based methodology delivered through the Kodacy virtual internship platform, conducted in association with SPACE under the AICTE Virtual Internship scheme, over a duration of 30 days. Learning was organized into structured modules, each covering a specific sub-topic within Artificial Intelligence and Machine Learning, accompanied by reading material, video content and short quizzes/assessments to reinforce understanding. Concepts learned were further reinforced through hands-on practice using Python and standard AI/ML libraries on sample datasets. On completion of all modules and the final assessment, a certificate of internship was issued jointly under Kodacy and SPACE, confirming successful completion of the 30-day “Artificial Intelligence and Machine Learning” internship programme.

#### **4.9 Expected Outcomes**

- A clear conceptual understanding of what Artificial Intelligence is and how it differs from traditional programming.

- Familiarity with core machine learning paradigms and common algorithms used in practice.
- Basic hands-on exposure to Python-based tools used for AI/ML development.
- Awareness of real-world AI applications and their impact across various industries.
- An appreciation of the ethical and societal considerations involved in building and deploying AI systems.
- A strong foundation to pursue further, more advanced studies and projects in AI and Machine Learning.
- Ability to understand and interpret basic AI and machine learning concepts, terminology, and workflows.
- Develop foundational skills in data collection, preprocessing, visualization, and preparation for machine learning tasks.
- Gain practical experience in developing and evaluating simple machine learning models using Python.
- Understand the importance of training, testing, validation, and performance evaluation of AI models.
- Develop problem-solving and analytical thinking skills through AI/ML-based projects and case studies.

#### **4.10 Certificate of Completion**

The certificate of internship issued by Kodacy, in association with SPACE, for successfully completing the 30-day virtual internship programme on “Artificial Intelligence and Machine Learning,” in the name of Arvinder Singh Bagga, dated September 8, 2026 (Certificate ID: 644b2875e2f9e315), has been attached in Chapter 3 of this report as proof of completion of the internship.

## CHAPTER 5: BIBLIOGRAPHY/REFERENCES

1. Russell, S., & Norvig, P. (2020). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.
2. Kodacy, in association with SPACE. (2026). Artificial Intelligence and Machine Learning – Internship Course Modules. Retrieved from Kodacy virtual internship platform.
3. AICTE. (2026). AICTE Internship Policy and Guidelines. All India Council for Technical Education.
4. Scikit-learn Developers. Scikit-learn: Machine Learning in Python. Retrieved from <https://scikit-learn.org>
5. Python Software Foundation. Python Language Reference. Retrieved from <https://www.python.org>
6. Murphy, K. P. (2022). *Probabilistic Machine Learning: An Introduction*. MIT Press